takes care of connecting with external data centre resources, such as Amazon Web Services, Eucalyptus, or OpenStack cloud.

The openQRM cloud portal provides a Web interface that internal or external users can access to compile IT resources, as needed.

Features

- Supports P2V, P2P, V2P, V2V migrations and high availability.
- Integrates with all major open and commercial storage technologies.
- Integrated billing system that maps CCU/h (cloud computing units) to real currency.
- Self-service portal for end users provisions new servers and application stacks in minutes!

Official website: *http://openqrm-enterprise.com*

OpenNebula

OpenNebula is a simple yet powerful and flexible turnkey open source solution to build private clouds and manage data centre virtualisation. The OpenNebula platform manages a data centre's virtual infrastructure to build private, public and hybrid implementations of Infrastructure as a Service. The two primary uses of the OpenNebula platform are data centre virtualisation solutions and cloud infrastructure solutions.

OpenNebula was designed to help companies build simple, cost-effective, reliable, open enterprise clouds on existing IT infrastructure. It provides flexible tools that orchestrate storage, network and virtualisation technologies to enable the dynamic placement of services. The design of OpenNebula is flexible and modular, to allow integration with different storage and network infrastructure and hypervisor technologies.

OpenNebula components include the following three layers:

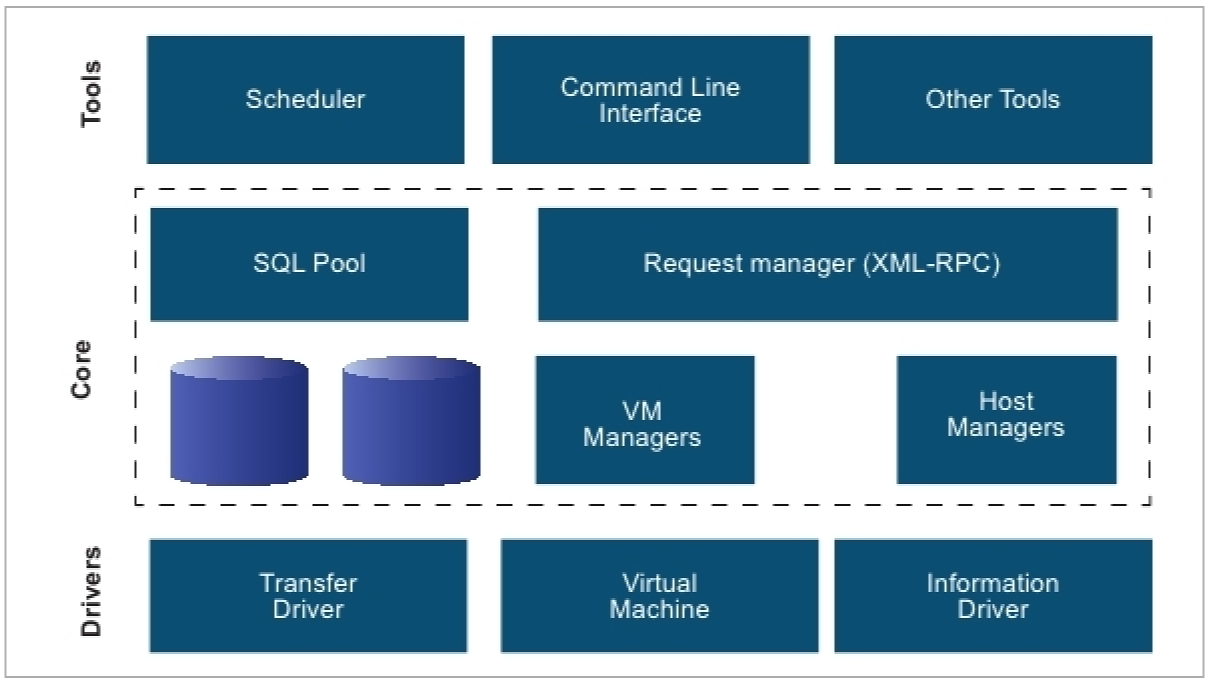

Figure 3: OpenNebula components

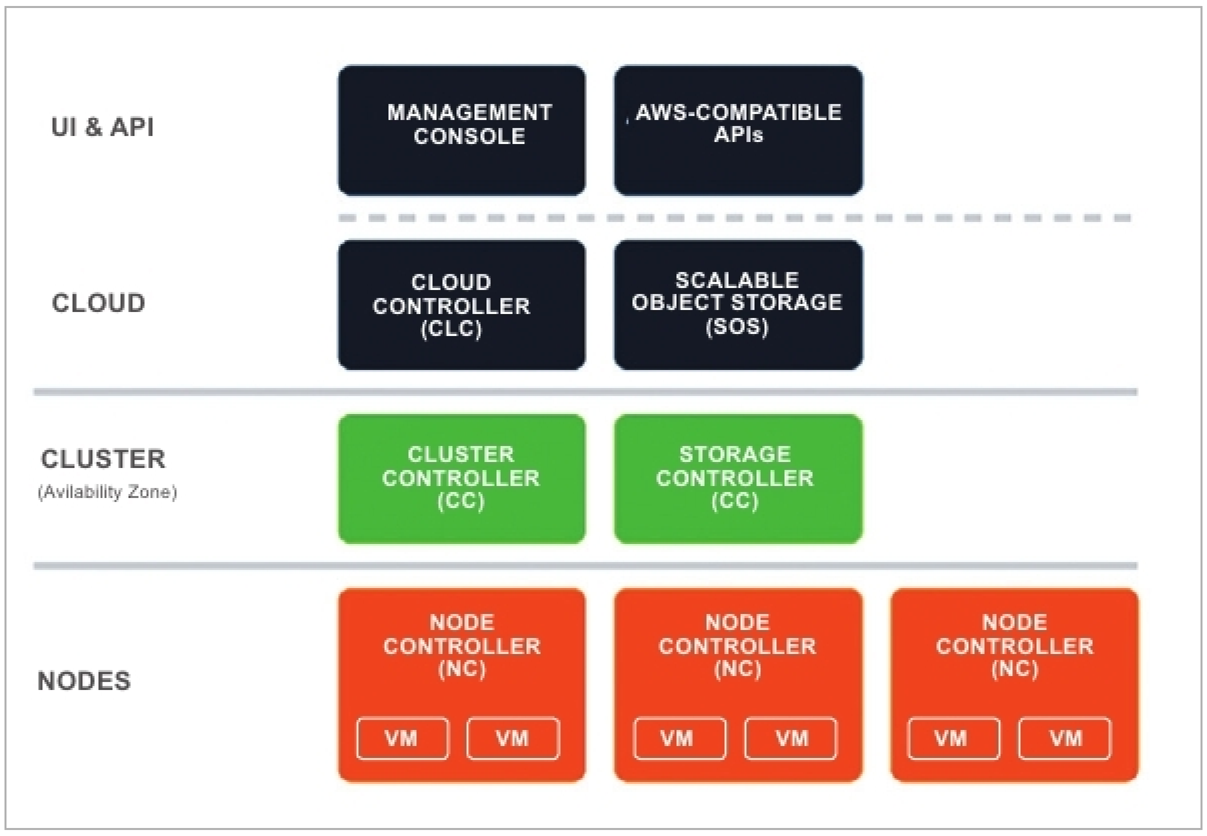

Figure 4: Eucalyptus architecture

1. The driver layer is responsible for the creation, start-up and shutdown of virtual machines (VMs), for allocating storage to VMs, and for monitoring the operational status of physical machines (PMs).
2. The core layer manages the VMs' full life cycle, including setting up virtual networks dynamically, dynamic IP address allocation for VMs and managing VMs' storage.
3. The tool layer provides interfaces, such as the command line interface (CLI), to communicate with users.

Features

- Supports numerous APIs like AWS EC2, EBS and OGF OCCI.
- Powerful UNIX based CLI for administration.
- GUI for cloud customers and data centre professionals.
- Resource allocation via fine-grained ACLs; load balancing, high availability, high performance computing.
- Powerful scheduling for task management.
- Supports integration with LDAP and Active directory.